MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, POWER
TYPES 2N1042 THROUGH 2N1045

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, germanium, power, transistors.

* 1.2 Physical dimensions. See figure 1.

* 1.3 Maximum ratings.

P_T 1/ T_C = 25°C	V_{CBO}				V_{EBO}	I_C	T_{stg} and T_{op}
	2N1042	2N1043	2N1044	2N1045			
W	Vdc	Vdc	Vdc	Vdc	Vdc	Adc	°C
20	-40	-60	-80	-100	-20	-3	-55 to +100

1/ Derate linearly 267 mW/°C for T_C > 25°C.

* 1.4 Primary electrical characteristics at T_C = 25°C ±3°C.

Limits	h_{FE} V_{CE} = -1 Vdc I_C = -3 Adc	h_{FE} V_{CE} = -0.5 Vdc I_C = -50 mAdc	$V_{CE}(sat)$ I_C = -3 Adc I_B = -300 mAdc	$\lvert h_{fe} \rvert$ V_{CE} = -1.5 Vdc I_C = -0.5 Adc f = 125 kHz	V_{BE} V_{CE} = -1 Vdc I_C = -3 Adc
			Vdc		Vdc
Min	20	50	---	2	---
Max	60	250	-0.75	10	-1.5

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

FSC 5961

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STANDARDS

MILITARY

* MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

* 3.3.1 Lead material and finish. Lead material shall be Kovar or alloy 52. Lead finish shall be gold-plated. (Leads may be tin-plated if specified in the contract or order, and this requirement shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking (see 6.2).

* 3.3.1.1 Lead material. If lead material need be specified, it shall be specified in the contract or order (see 6.2).

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

* 3.5 Marking. Devices shall be marked in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

* 4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

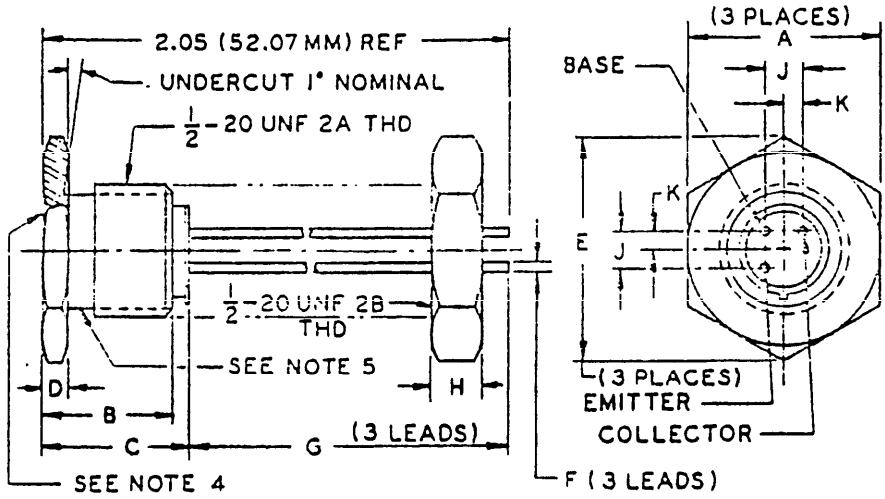
* 4.3 Quality conformance inspection. Quality conformance inspection shall consist of group A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

* 4.3.3 Group C inspection. Group C inspection shall consist of the tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every six months during production.

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.740	.760	18.80	19.30	6
B	.485	.515	12.32	13.08	
C	---	.550	---	13.97	
D	.083	.103	2.11	2.62	
E	---	.875	---	22.23	6
F	.022	.028	.56	.71	7
G	1.500	1.750	38.10	44.45	7
H	.177	.197	4.50	5.00	
J	.1314	.1514	3.34	3.85	
K	.0707 Nom		1.80 Nom		

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. The orientation of the leads in relation to the hex flats is not controlled.
3. The collector shall be electrically connected to the case.
4. Temperature measurement point 0.160 nominal from center of heat sink.
5. Thread relief is 0.090 max. by 0.430 dia. nominal.
6. Three places.
7. Three leads.

FIGURE 1. Physical dimensions of transistor types 2N1042 through 2N1045.

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TABLE I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details (see 4.4.3)			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Breakdown voltage, collector to base	3001	Bias condition D: I_C = -750 μAdc					
2N1042				BV_{CBO}	-40	---	Vdc
2N1043					-60	---	Vdc
2N1044					-80	---	Vdc
2N1045					-100	---	Vdc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = -100 mAdc					
2N1042				BV_{CEO}	-30	---	Vdc
2N1043					-40	---	Vdc
2N1044					-50	---	Vdc
2N1045					-60	---	Vdc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = -20 Vdc		I_{EBO}	---	-650	μAdc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = +0.2 Vdc					
2N1042		V_{CE} = -40 Vdc		I_{CEX}	---	-650	μAdc
2N1043		V_{CE} = -60 Vdc			---	-650	μAdc
2N1044		V_{CE} = -80 Vdc			---	-650	μAdc
2N1045		V_{CE} = -100 Vdc			---	-650	μAdc
Collector to base cutoff current	3036	Bias condition D					
2N1042		V_{CB} = -20 Vdc		I_{CBO}	---	-125	μAdc
2N1043		V_{CB} = -30 Vdc			---	-125	μAdc
2N1044		V_{CB} = -40 Vdc			---	-125	μAdc
2N1045		V_{CB} = -50 Vdc			---	-125	μAdc
Collector to emitter cutoff current	3041	Bias condition D					
2N1042		V_{CE} = -15 Vdc		I_{CEO}	---	-25	mAdc
2N1043		V_{CE} = -20 Vdc			---	-20	mAdc
2N1044		V_{CE} = -25 Vdc			---	-20	mAdc
2N1045		V_{CE} = -30 Vdc			---	-20	mAdc
* Subgroup 3			5				
Forward-current transfer ratio	3076	V_{CE} = -1 Vdc; I_C = -3 Adc pulsed (see 4.4.1)		h_{FE}	20	60	---
Forward-current transfer ratio	3076	V_{CE} = -1 Vdc; I_C = -1 Adc pulsed (see 4.4.1)		h_{FE}	30	150	---
Forward-current transfer ratio	3076	V_{CE} = -0.5 Vdc; I_C = -50 mAdc		h_{FE}	50	250	---

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TABLE I. Group A inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details (see 4.4.3)			Min	Max	
Subgroup 3 - continued							
Collector to emitter voltage (saturated)	3071	I_C = -3 Adc; I_B = -300 mAdc pulsed (see 4.4.1)		$V_{CE}(sat)$	---	-0.75	Vdc
Collector to emitter voltage (saturated)	3071	I_C = -1 Adc: I_B = -100 mAdc pulsed (see 4.4.1)		$V_{CE}(sat)$	---	-0.25	Vdc
Base emitter voltage (nonsaturated)	3066	Test condition B; V_{CE} = -1 Vdc; I_C = -3 Adc pulsed (see 4.4.1)		V_{BE}	---	-1.5	Vdc
Base emitter voltage (nonsaturated)	3066	Test condition B; V_{CE} = -1 Vdc; I_C = -50 mAdc		V_{BE}	---	-0.5	Vdc
Subgroup 4			5				
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -1.5 Vdc; I_C = -0.5 Adc		h_{fe}	25	100	---
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = -1.5 Vdc; I_C = -0.5 Adc; f = 125 kHz		$\lvert h_{fe} \rvert$	2	10	---
Subgroup 5			10				
High-temperature operation:		T_C = +85°C					
Collector to base cutoff current	3036	Bias condition D					
2N1042		V_{CB} = -20 Vdc		I_{CBO}	---	-5	mAdc
2N1043		V_{CB} = -30 Vdc			---	-5	mAdc
2N1044		V_{CB} = -40 Vdc			---	-5	mAdc
2N1045		V_{CB} = -50 Vdc			---	-5	mAdc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = +0.2 Vdc					
2N1042		V_{CE} = -20 Vdc		I_{CEX}	---	-5	mAdc
2N1043		V_{CE} = -30 Vdc			---	-5	mAdc
2N1044		V_{CE} = -40 Vdc			---	-5	mAdc
2N1045		V_{CE} = -50 Vdc			---	-5	mAdc
Forward-current transfer ratio	3076	V_{CE} = -1 Vdc; I_C = -3 Adc pulsed (see 4.4.1)		h_{FE}	20	75	---
Low-temperature operation:		T_C = -55°C					
Forward-current transfer ratio	3076	V_{CE} = -1 Vdc; I_C = -3 Adc pulsed (see 4.4.1)		h_{FE}	15	---	---

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TABLE II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details (see 4.4.3)			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
* Subgroup 2			15				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition A, except in step 3, T_A = +100° +5, -0°C		---	---	---	---
Thermal shock (glass strain)	1056	Test condition A		---	---	---	---
Hermetic seal	1071	Test condition G or H for fine leaks; test condition A, C, D or F for gross leaks		---	---	$5x10^{-7}$	atm cc/s
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias condition D					
2N1042		V_{CB} = -20 Vdc		I_{CBO}	---	-125	µAdc
2N1043		V_{CB} = -30 Vdc			---	-125	µAdc
2N1044		V_{CB} = -40 Vdc			---	-125	µAdc
2N1045		V_{CB} = -50 Vdc			---	-125	µAdc
Forward-current transfer ratio	3076	V_{CE} = -1 Vdc; I_C = -3 Adc pulsed (see 4.4.1)		h_{FE}	20	60	---
* Subgroup 3			10				
Shock	2016	Nonoperating; 1,500 G; 0.5 ms; 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration, variable frequency	2056			---	---	---	---
Constant acceleration	2006	10,000 G in each orientation: X_1, Y_1, Y_2 and Z_1					
End points: (Same as subgroup 2)							
* Subgroup 4			10				
Terminal strength (lead fatigue)	2036	Test condition E		---	---	---	---
Terminal strength (stud torque)	2036	Test condition D2; torque = 20 in-lbs; time = 15 s		---	---	---	---

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TABLE II. Group B inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details (see 4.4.3)			Min	Max	
Subgroup 4 - Continued							
End points:							
Hermetic seal	1071	Test condition G or H for fine leaks; test condition A, C, D or F for gross leaks		---	---	$5x10^{-7}$	atm cc/s
* Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
* Subgroup 6			7				
High-temperature life (nonoperating)	1032	T_{stg} = +100°C; time = 340 hours (see 4.3.4)		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias condition D					
2N1042		V_{CB} = -20 Vdc		I_{CBO}	---	-250	µAdc
2N1043		V_{CB} = -30 Vdc			---	-250	µAdc
2N1044		V_{CB} = -40 Vdc			---	-250	µAdc
2N1045		V_{CB} = -50 Vdc			---	-250	µAdc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = +0.2 Vdc					
2N1042		V_{CE} = -40 Vdc		I_{CEX}	---	-1.3	mAdc
2N1043		V_{CE} = -60 Vdc			---	-1.3	mAdc
2N1044		V_{CE} = -80 Vdc			---	-1.3	mAdc
2N1045		V_{CE} = -100 Vdc			---	-1.3	mAdc
Forward-current transfer ratio	3076	V_{CE} = -1 Vdc; I_C = -3 Adc pulsed (see 4.4.1)		h_{FE}	16	72	---
* Subgroup 7			7				
Steady-state operation life	1027	+25°C ≤ T_C ≤ +55°C; V_{CE} = -20 Vdc; $P_T = 12W + \left(\frac{55°C - T_C}{3.75°C/W}\right)$; time = 340 hours (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							

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TABLE III. Group C inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details (see 4.4.3)			Min	Max	
Subgroup 1			10				
Thermal resistance (junction to case)	3136	$T_1 = T_C = 30 \pm 5°C$ $T_2 = T_J = 95 \pm 5°C$ I_C (measurement) = -50 mAdc		θ_{J-C}	---	3.75	°C/W
* Subgroup 2			10				
Resistance to solvents	---	MIL-STD-202, Method 215 (see 4.4.2)		---	---	---	---
* Subgroup 3			$\lambda = 10$				
High-temperature life (nonoperating)	1031	$T_{stg} = +100°C$ (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							
* Subgroup 4			$\lambda = 10$				
Steady-state operation life	1026	$+25°C \leq T_C \leq +55°C$ $V_{CE} = -20$ Vdc $P_T = 12W + \left(\frac{55°C - T_C}{3.75°C/W}\right)$ (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							

* 4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hour life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria, see 4.3.3.

* 4.3.5 Lot representative (group B and C inspections). At the option of the manufacturer, the highest voltage type represented in the lot may be used for group B and C inspections as representative of a lot containing the several types.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III.

* 4.4.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

* 4.4.2 Resistance to solvents. Transistors shall be subjected to tests in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the transistor body where marking has been applied shall be brushed.
(b) After subjection to the tests there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

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* 4.4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be conducted at a case temperature (T_C) of 25°C ±3°C.

5. PREPARATION FOR DELIVERY

* 5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

* 6.2 Ordering data.

(a) Lead finish if other than gold-plated (see 3.3.1).
(b) Lead material (see 3.3.1.1).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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